Rindania kaltsum
201331056
2B - TTE

No.
Date 25 . 03 . 22

UTS Elkom.

1. Dik : Rangkaian Resonansi LC paralel
L = 39,6 μH
C = 1 nF
Frek : 850 kHz
GLC Linier : Q = 100
2 isyarat, amplitudo = 100 μ A
Frek berbeda : 850 kHz dan 900 kHz = Masuk antena

Dit : Hitung besar tegangan yang muncul pada rangkaian penala tsb ?

Jaw : Q = 100

$$G = \frac{100 \times C}{Q} = \frac{2\pi . 850 . 10^3 . 10^{-9}}{100} = 0{,}0534 \text{ mU}$$

$$R = \frac{1}{G} = \frac{1}{0{,}0534} = 18{,}72 \text{ k}\Omega$$

Isyarat pertama :

$$V_1 = I_1 R = 100 \cdot 10^{-6} \; 18{,}72 . 10^3 = 1{,}872 \text{ volt}$$

Isyarat kedua :

$$y_2 = \sqrt{G^2 + (\omega . C . {}^1/_{\omega L})^2}$$

$$y_2 = \sqrt{G^2 + \left(2\pi f C - \frac{1}{2\pi f L}\right)^2}$$

$$y_2 = \sqrt{(0{,}0534 . 10^{-3})^2 + \left(2\pi . 900 . 10^3 . 10^{-9} - \frac{1}{2\pi . 900 . 10^3 . 39{,}6 . 10^{-6}}\right)}$$

$$= 0{,}001190 = 0{,}0012 \text{ mU}$$

Isyarat kedua akan terukur tegangan

$$V_2 = \frac{I_2}{Y_2} = \frac{100 . 10^{-6}}{0{,}0012 . 10^{-3}} = 0{,}083 = 83{,}3 \text{ mV}$$

$\therefore V_1 = 1{,}872$ volt //

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2. Dik: type L, sumber DC

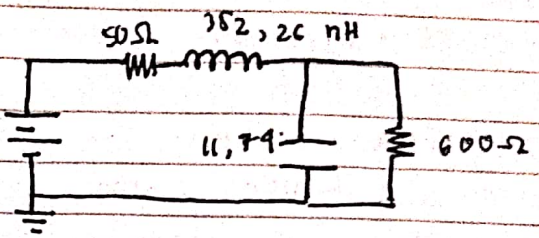
$R_s = 50\ \Omega$

$R_L = 600\ \Omega$

$f = 75\ MHz$

$Q_s = Q_p = \sqrt{\frac{R_L}{R_s} - 1} = \sqrt{\frac{600}{50} - 1} = 3,32$

Rangkaian



$X_s = Q_s R_s = 3,32\ (50) = 166\ \Omega$

$X_p = \frac{R_L}{Q} = \frac{600}{3,32} = 180,72\ \Omega$

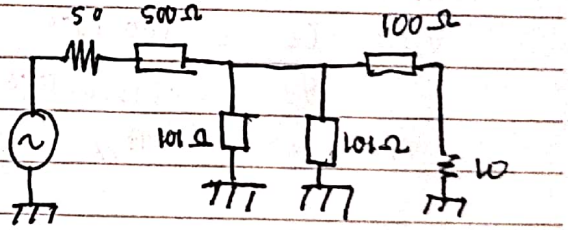
$L = \frac{X_s}{2\pi F} = \frac{166}{2\pi (75\ MHz)} = 352,26\ nH$

$C = \frac{1}{2\pi f X_p} = \frac{1}{2\pi (75\ MHz)(180,72)} = 11,79\ pH$

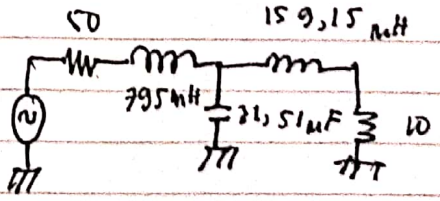
3. $Q = \sqrt{\frac{R}{R_s} - 1} \rightarrow R = R_{kecil}\ (Q^2 + 1) = (10)(101) = 1010\ \Omega$

- $X_{s1} = Q(R_s) = Q(R) = 10\ (50) = 500\ \Omega$
- $X_{pi} = \frac{R_p}{Q_p} = \frac{R}{Q} = \frac{1010}{10} = 101\ \Omega$
- Rangkaian L untuk beban

$Q = \sqrt{\frac{R}{R_{p2}=R_L} - 1} = \sqrt{\frac{1010}{10} - 1} = 10$



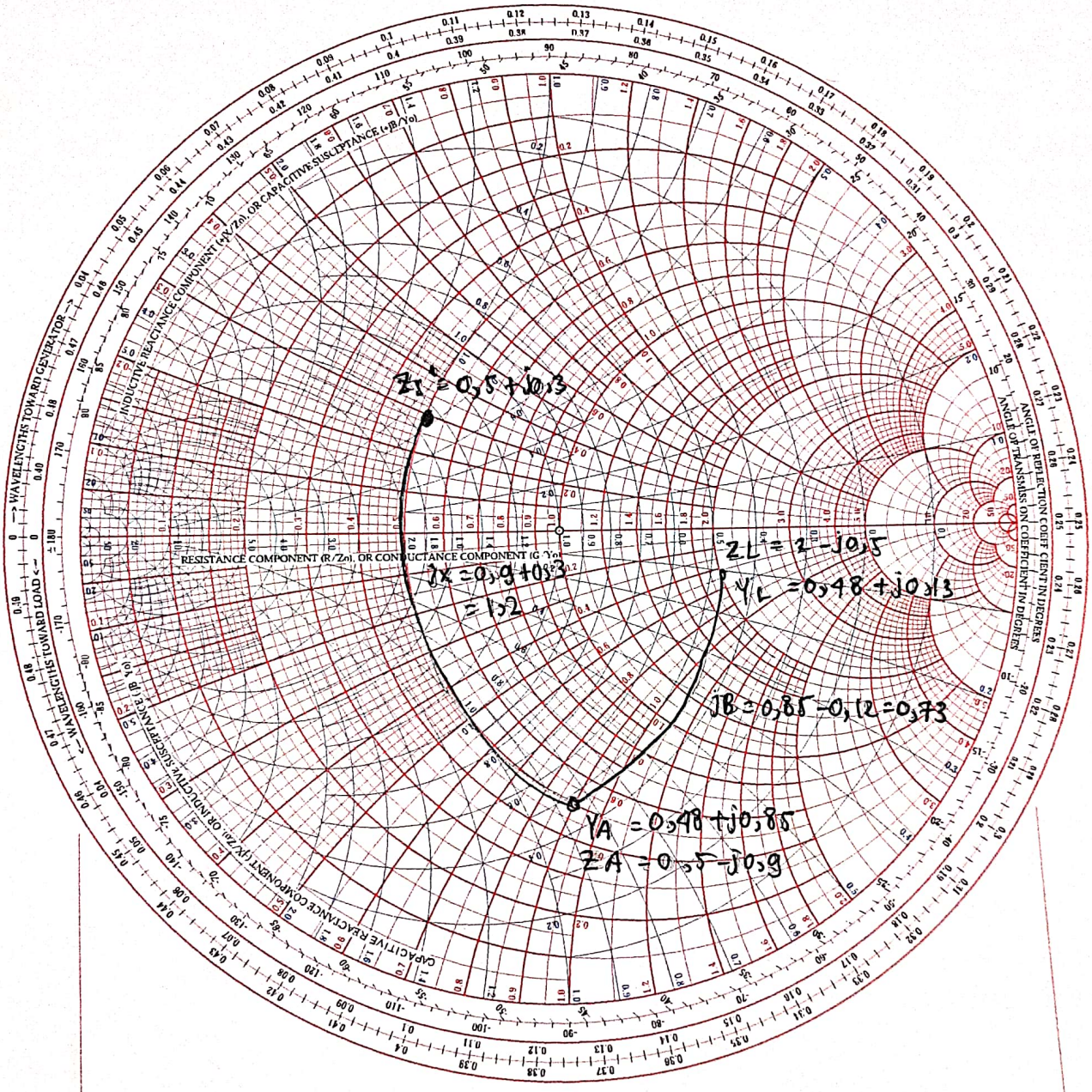
- $X_{p2} = \frac{R_p}{Q_2} = \frac{R}{Q} = \frac{1010}{10} = 101\ \Omega$
- $X_{s2} = Q_2(R_L) = Q(R_L) = 10\ (10) = 100\ \Omega$
- $X_{ptot} = X_{p1} // X_{p2} = 101 // 101 = 50,5\ \Omega$
- $C = \frac{1}{2\pi f (X_{ptot})} = \frac{1}{2\pi (100\ kHz)\, 50,5} = 31,51\ nF$
- $L_1 = \frac{X_{s1}}{2\pi f} = \frac{500}{2\pi (100\ kHz)} = 795\ mH$
- $L_2 = \frac{X_{s2}}{2\pi f} = \frac{100}{2\pi (100\ kHz)} = 159,15\ mH$

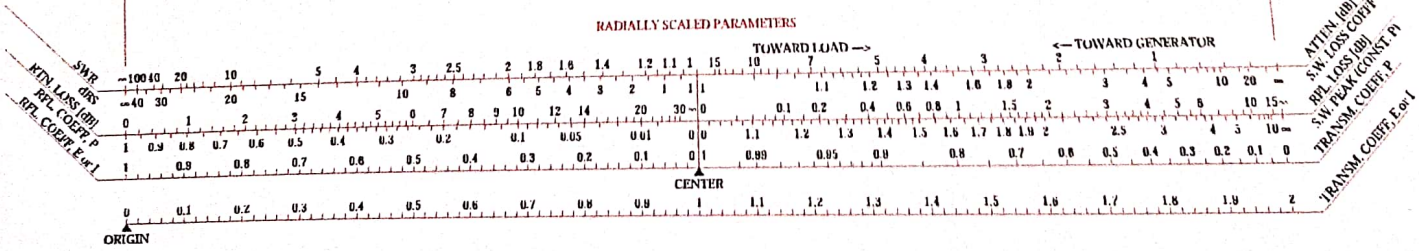


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Rindania kaltsum	NO. 4	
SMITH CHART	Courtesy of Microwaves101.com	DATE 25 - 03 - 2022

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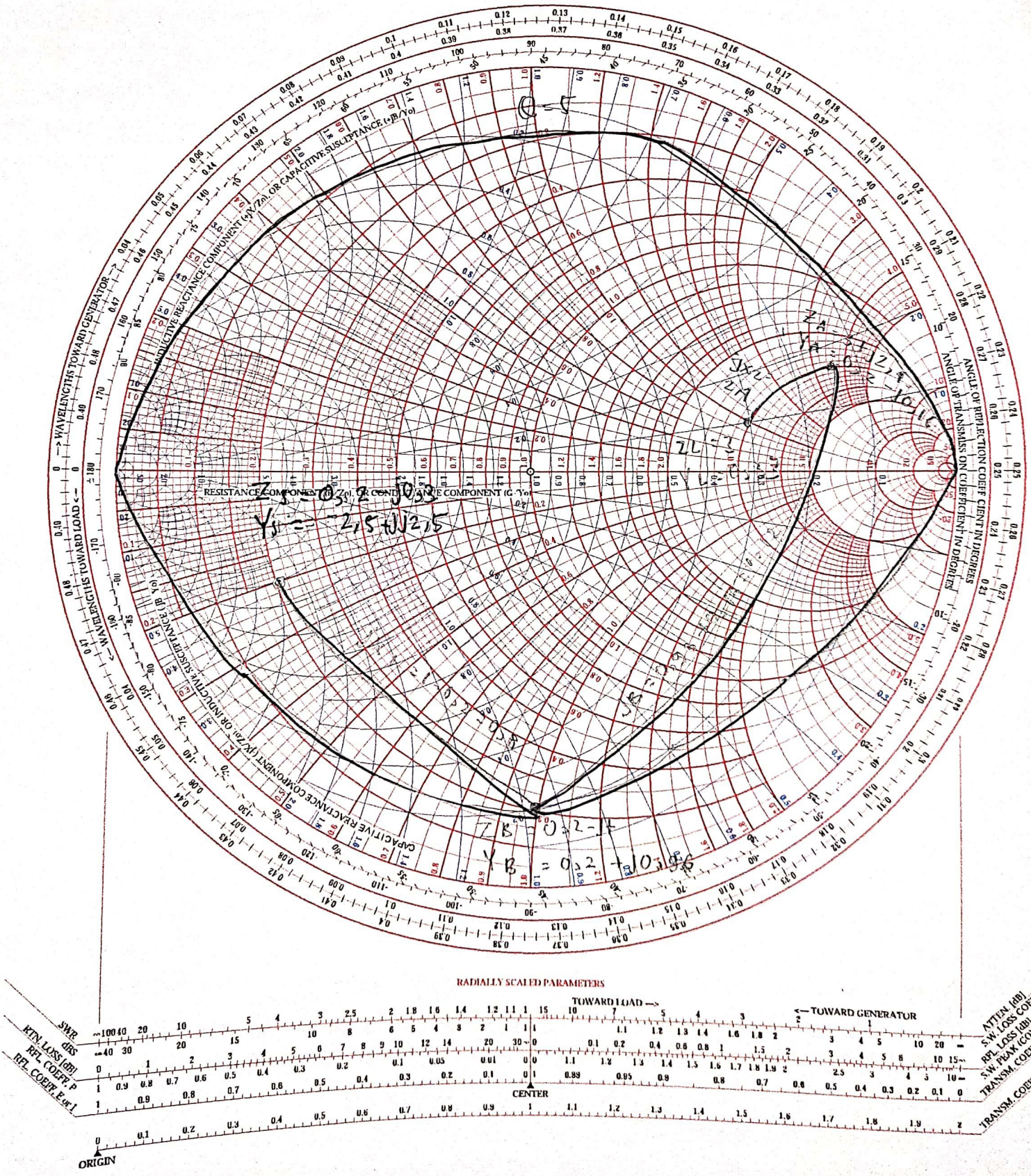




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